JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY GURAJADA VIZIANAGARAM

SRI SIVANI COLLEGE OF ENGINEERING & EXAM CELL

DEPARTMENT OF ELECTRICAL & ELECTRONICS ENGINEERING

★ PREVIOUS SEMESTER QUESTION PAPERS WITH SOLVED DERIVATIONS ★

Electrical Machines-I & Magnetic Circuits Previous Semester Question Papers

Course Code:	R23-EEE-2101	**Academic Scheme:**	II B.Tech I Semester (R23 Regulation)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Paper 1: Electromechanical Energy Conversion Principles

Forces and torques in magnetic field systems, Energy and co-energy concepts, Singly excited and doubly excited magnetic systems.

Paper 2: DC Generators Principles & Armature Reaction

Construction details, Lap and wave windings, EMF equation, Armature reaction, Demagnetizing and cross-magnetizing amp-turns, Commutation and interpoles.

Paper 3: DC Motors & Operating Characteristics

Principle of operation, Back EMF, Torque equation, Characteristics of shunt, series, and compound motors, Applications.

Paper 4: Speed Control & Starting of DC Motors

Armature resistance control, Field flux control, Ward-Leonard system, 3-point and 4-point starters, Calculation of starter resistance steps.

Paper 5: Testing of DC Machines & Efficiency Analysis

Losses in DC machines, Brake test, Swinburne's test, Hopkinson's regenerative test, Retardation test, Separation of core losses.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Exam Pattern	Part A: 10 Short questions (20M) + Part B: 5 Units essay (50M)
Total Marks	70 Marks
Prescribed Textbook	Electrical Machines by P.S. Bimbhra / I.J. Nagrath & D.P. Kothari

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